

Algebraic manipulation:

Transposing, expanding and factorising

Key topics

Transposing equations

Expanding brackets

Factorising expressions

Algebra

Algebra is the area of Mathematics where we **use letters to represent missing or unknown values**

It is very powerful for problem solving, especially when you have some but not all of the information needed.

The golden rule is that whatever you do to one side of the equation, you also do to the other side.

Letters as numbers

There are lots of conventions for which symbols we pick:

$\alpha\beta\gamma$ – radiation!

δ – small changes (differentiation)

λ – wavelength/eigenvalues

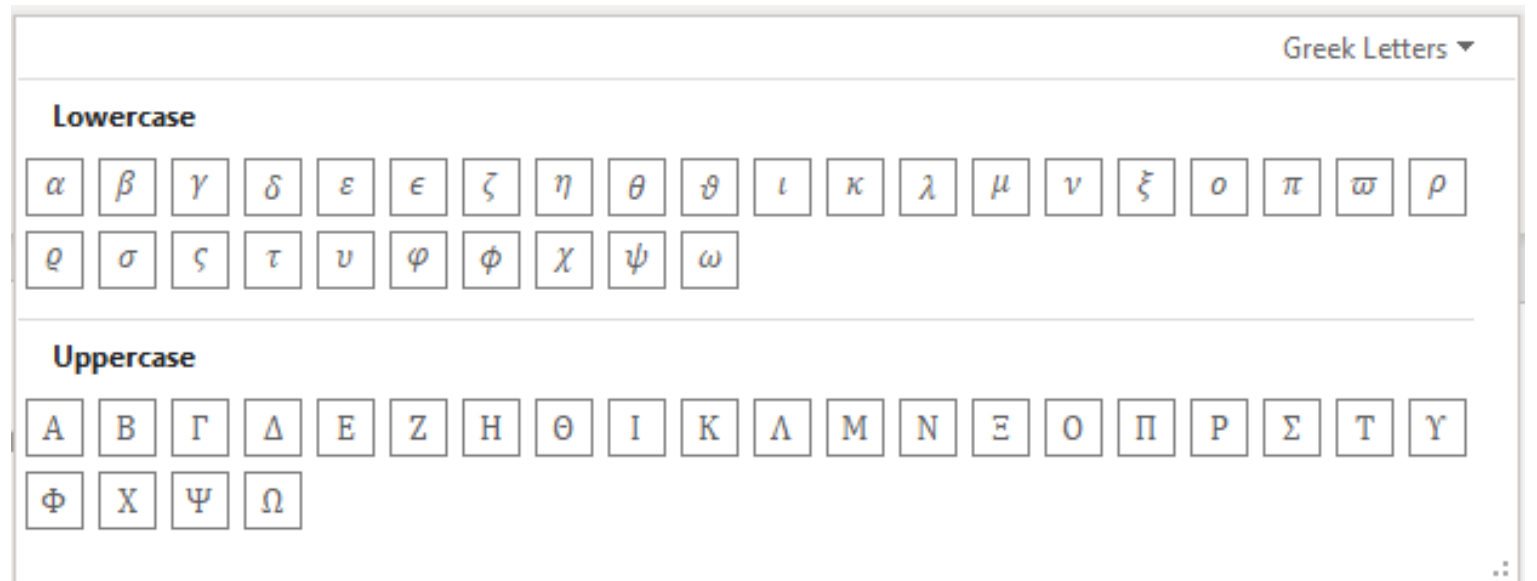
$\mu\sigma\chi$ - statistics

$\theta\phi$ - angles

π constant

ε – errors

+ many more...



Algebraic rearrangement - transposition

Starting with the simple equation:

$$3 + x = 7$$

We can deduct three from each side:

$$x = 7 - 3, \quad x = 4$$

We have made x the subject of the equation by isolating the x terms on the left hand side

Simple Examples

$$2x + 3 = 15$$

$$2x = 15 - 3$$

$$x = \frac{15 - 3}{2}$$

$$x = 6$$

$$2x + 2 = 11 - x$$

$$2x + 2 + x = 11$$

$$2x + x = 11 - 2$$

$$3x = 9$$

$$x = 3$$

Isolate x

Move all x terms to one side

Move all non x terms to the other side

Divide or multiply as needed to find one unit of x (i.e. not $2x$ or $3x$)

$$y = mx + c$$

$$y - c = mx$$

$$\frac{y - c}{m} = x$$

$$x = \frac{y - c}{m}$$

Common mistakes (1)

Dividing by x

$$2(x + 3) = 16$$

$$2(1 + 3) = \frac{16}{x}$$

Addition Vs
Multiplication

$$2x + x = 2x^2$$

$$2x \times x = 3x$$

Combining terms

$$2x + 2y = 2xy$$

$$2x + 2x^2 = 4x^2$$

Common mistakes (2)

$$\sqrt{a^2 + b^2} \neq a + b$$

$$\frac{a}{b + c} \neq \frac{a}{b} + \frac{a}{c}$$

Further Examples

$$T = 2\pi \sqrt{\frac{l}{g}}, \text{ solve for } l$$

$$\frac{T}{2\pi} = \sqrt{\frac{l}{g}}$$

$$\left(\frac{T}{2\pi}\right)^2 = \frac{l}{g}$$

$$l = \frac{T^2 g}{4\pi^2}$$

$$\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}$$

$$\frac{1}{x} = \frac{2}{y} - \frac{3}{z}, \text{ solve for } z$$

$$\frac{1}{x} + \frac{3}{z} = \frac{2}{y}$$

$$\frac{3}{z} = \frac{2}{y} - \frac{1}{x}$$

$$3\frac{1}{z} = \frac{2}{y} - \frac{1}{x}$$

$$\frac{1}{z} = \frac{1}{3} \left(\frac{2}{y} - \frac{1}{x} \right)$$

$$\frac{1}{z} = \frac{1}{3} \left(\frac{2x - y}{xy} \right)$$

$$z = 3 \left(\frac{xy}{2x - y} \right) = \frac{3xy}{2x - y}$$

Expanding brackets

$$(2 + 3) \times (4 + 5) \qquad (5) \times (9) \qquad 45$$

Everything in the first bracket is multiplied by everything in the second

$$(2 + 3) \times (4 + 5) \qquad 2 \times 4 + 2 \times 5 + 3 \times 4 + 3 \times 5 \qquad 8 + 10 + 12 + 15 = 45$$

This seems more clumsy, but it gets the answer right and is easier to use with letters

$$(a + b) \times (c + d) \qquad ac + ad + bc + bd \qquad a(c + d) + b(c + d)$$

We can expand the idea to multiple brackets, but it gets complicated quite fast...

Big example!

$$(x + 1)^3 = (x + 1)(x + 1)(x + 1)$$

$$(x + 1)[(\mathbf{x + 1})(\mathbf{x + 1})] = (x + 1)[\mathbf{x^2 + 2x + 1}]$$

$$(x + 1)[\mathbf{x^2 + 2x + 1}] = x^3 + 2x^2 + x^2 + x + x^2 + 2x + 1$$

$$(x + 1)^3 = x^3 + 4x^2 + 3x + 1$$

MASSIVE example!

$$(x + 4)(x + 3)(x + 2)(x + 1)$$

$$(x + 2)(x + 1) = x^2 + 2x + x + 2 = x^2 + 3x + 2$$

$$(x + 4)(x + 3) = x^2 + 3x + 4x + 12 = x^2 + 7x + 12$$

$$(x^2 + 7x + 12)(x^2 + 3x + 2)$$

$$= x^4 + 3x^3 + 2x^2 + 7x^3 + 21x^2 + 14x + 21x^2 + 36x + 24$$

$$= x^4 + 10x^3 + 44x^2 + 50x + 24$$

Factorising

Some expressions have common factors which we can take out

$$3x + 4qx + 12wx =$$
$$x(3 + 4q + 12w)$$

This also works with fractions:

$$\frac{2}{7} + \frac{2}{9} = 2\left(\frac{1}{7} + \frac{1}{9}\right)$$

Factorising usually makes an equations simpler (easier to write) It can be useful for isolating a variable or finding zeros of functions

Example

$$3x^2 + 6x$$

We have two terms: $3x^2$ and $6x$

These terms have some common factors, both involve multiples of 3 and both involve x .

$$3x^2 + 6x = 3x(x + 2)$$

Expanded form = factorised form

Factorising to help rearrange expressions

Rearrange this expression to make x the subject

$$3x + xy = z$$

$$x(3 + y) = z$$

$$x = \frac{z}{3 + y}$$

Factorising expressions

$$x^2 + 4x + 3$$

This expression combines two factors, we can see this if we rewrite it like this:

$$(x + 1)(x + 3)$$

$(x + 1)$ is a factor and so is $(x + 3)$

The two x terms are multiplied to give us x^2

The 1 and 3 are each multiplied by x , giving us: $x+3x=4x$

The 1 and 3 are multiplied together to give us 3

Process

$$x^2 + 4x + 3$$

How do we pick the values?

$$(? + ?)(? + ?)$$

We need to get x^2 so we must have $(x + ?)(x + ?)$

The numbers are ones that **add** to give 4 and **multiply** to give 3

$$3 + 1 = 4, 3 \times 1 = 3$$

$$x^2 + 4x + 3 = (x + 3)(x + 1)$$

Common factors in terms

$$2x^2 + 8x + 6$$

This is an example where we can start by taking out a common factor from all terms (2), giving us:

$$2(x^2 + 4x + 3)$$

Which is the same example we just had! If you can simplify an expression before factorising it, do so

$$2(x^2 + 4x + 3) = 2(x + 3)(x + 1)$$

Multiple solutions

$$2x^2 + 5x + 3$$

We have a $2x^2$ term again and taking 2 out as a common factor gives:

$$2(x^2 + 2.5x + 1.5)$$

1.5 and 1 add to give 2.5 and multiply to give 1.5 so a valid solution is:

$$2(x + 1.5)(x + 1)$$

But this isn't the only solution!

$$(2x + 3)(x + 1)$$

Tricky factorising examples

$$(x + y)^3 + 7(x + y)^2$$

$$(x + y)^2(x + y + 7)$$

$$(x^2 + 2y)^2 + 2x^3 + 4xy$$

$$(x^2 + 2y)^2 + (x^2 + 2y)2x$$

$$(x^2 + 2y)(x^2 + 2y + 2x)$$

Difference of two squares

Some results are well known because they crop up a lot and are Mathematically 'neat'.

$$(a - b)(a + b)$$

$$a^2 + ab - ba - b^2$$

$$a^2 - b^2$$

$$(a - b)(a + b) = a^2 - b^2$$

This is called the difference of two squares

Example of difference of two squares

Expand: $x^2 - 4y^2$

Difference of two squares gives: $a^2 - b^2 = (a - b)(a + b)$

We have: $x^2 - 4y^2 = x^2 - (2y)^2$

So: $x^2 - 4y^2 = (x - 2y)(x + 2y)$

Perfect squares

Another two important expressions are:

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

These are called perfect squares

Example of perfect squares

Expand $(2x - 3y)^2$

We have the result:

$$(a - b)^2 = a^2 - 2ab + b^2$$

We replace a by $2x$ and b by $3y$:

$$(2x - 3y)^2 = (2x)^2 - 2(2x)(3y) + (3y)^2$$

$$(2x - 3y)^2 = 4x^2 - 12xy + 9y^2$$

Cubes

It is also true of cubes, though we will generally use them less than the previous results:

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Summary

Different ways of expressing the same equation are possible.

Being able to move between them is an important skill.

Later in the course we'll look at using these to find out information about functions.